

PERRY CHEUNG

Senior Frontend Engineer | AI-Augmented Engineering | React / TypeScript, Next.js, Node.js

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Professional Summary

- **Senior Frontend Engineer** operating as a **high-leverage, AI-augmented team of one** — **architecting, building, and shipping** at the pace most companies need 3-4 engineers for. 10+ years delivering enterprise-scale **React/TypeScript** platforms **across FinTech, Retail POS SaaS, and GovTech**.
- 10+ years delivering **enterprise-scale React/TypeScript platforms** across **FinTech, SaaS, and GovTech**
- Sole architect of the enterprise **React** foundation at **Haitong International Securities** — built once, adopted across 5 platforms without modification, **still the reference architecture years later**
- **Led a 5-engineer frontend team** to **digitize 200+ enterprise trading workflows**; owned code review standards, architecture decisions, and mentorship
- Currently building **Sniff & Frolic** solo — **architecting and shipping** a production **AI-integrated POS + e-commerce platform (Next.js, NestJS, PostgreSQL, Claude API)**, with an eye toward **SaaS**.
- Full-stack capability in **Node.js** and **NestJS** to own **API contracts and data flow end-to-end, not just UI**

Technical Skills

Frontend Architecture: React, TypeScript, Next.js, Redux/Zustand, AgGrid, D3.js, Storybook, HTML5/CSS3 (SASS)

Backend & APIs: Node.js, NestJS, C# (.NET), API Design (REST/GraphQL), PostgreSQL, MongoDB

AI & Tools: Claude API (production integration, constrained context design), OpenAI API, LLM-assisted development workflows, prompt engineering for production reliability, Cursor

Cloud & DevOps: Azure, AWS (EC2/S3), Docker, Kubernetes, CI/CD (Jenkins, GitHub Actions), Git

Testing & Quality: Cypress (E2E), Jest, Unit & Integration Testing

UX & Accessibility: WCAG Accessibility Standards, Figma

Working Experience

Senior Frontend Engineer (UX/UI Developer)

Oct 2024 - Jan 2026

TechPOS · Canada Top Retail SaaS · 1,000+ dispensaries

Vancouver, BC, Canada

- **Legacy Modernization:** Owned the frontend modernisation strategy for an **8-year legacy POS platform** — designed an incremental migration path that delivered zero-downtime cutover with full **Cypress E2E** coverage protecting production.
- **Performance Engineering:** Architected a high-performance reporting engine handling 10,000+ records with zero UI lag, using AgGrid virtualisation and intelligent data summarisation — replacing a system merchants had stopped trusting.
- **Product Impact:** Designed and shipped a **Smart Setup onboarding** workflow with a predictive UI suggestion engine, **cutting merchant onboarding time by 40%** and **reducing support escalations**.
- **Next.js Migration:** Improved organic discoverability and achieved a **30% boost in page load speed** by orchestrating a legacy codebase revamp to **Next.js**, leveraging **SSR/SSG** and strategic code-splitting for **optimal SEO performance**.
- **AI-Assisted Delivery:** Integrated GitHub Copilot into daily engineering workflow to accelerate migration and feature delivery without compromising code review standards.

Stack: React, Next.js, TypeScript, Redux, Material UI, AgGrid, REST APIs, PostgreSQL, Storybook, Cypress, C#, Azure

Senior Frontend Engineer

Mar 2023 - Feb 2024

Radical I/O Company · Vancouver GovTech · SimpliCity CMS for BC municipalities

Vancouver, BC, Canada

- **Reusable Abstraction Engine:** Architected a declarative, **drag-and-drop JSON-driven form engine** as a reusable abstraction — supporting conditional logic, dynamic validation, and API integration — eliminating bespoke engineering work for each new form type and **reducing government permit processing time by 50%**.
- **Low-Code Empowerment:** Eliminated the need for engineering intervention in content updates by building **high-complexity CMS authoring tools**, enabling non-technical users to **manage dynamic layouts and media** independently
- **Scalable SaaS Architecture:** Engineered multi-tenant SaaS CMS platforms using **React** and **TypeScript**, implementing a modular architecture that maximized code reuse across enterprise projects.

Stack: React, TypeScript, Svelte, Node.js, PostgreSQL, MongoDB, REST APIs, Jest, Jenkins, AWS (EC2, S3), Git

Founder & Lead Engineer (Solo-built)

Feb 2024 – Present

Sniff & Frolic · Headless E-commerce platform · POS

Hong Kong, China / Vancouver (Remote)

- **End-to-End Product Ownership:** Owned full product design and engineering — from **UX flows in Figma** to **production implementation** — as sole designer and engineer across storefront, POS, and backend systems.
- **Headless Architecture:** Architected a **headless e-commerce platform** using **Next.js**, integrating WooCommerce APIs and transitioning to a custom **NestJS + PostgreSQL** backend to improve performance, data control, and system flexibility
- **Performance Engineering:** Designed product, cart, and checkout systems with **SSR/SSG optimization** for **performance** and **SEO**
- **POS System:** Built a **modular POS system** using **React, TypeScript**, and **Zustand** to support real-time sales operations and multi-role workflows
- **AI-Augmented Delivery:** Compressed a typical **3-4 people** build — **UX design, architecture, backend, frontend, AI integration** — into solo execution using AI-assisted development workflows across the stack.

Stack: React, Next.js, TypeScript, Zustand, Node.js (NestJS), REST APIs, PostgreSQL, Storybook, Playwright, Vercel, Supabase

Lead Frontend Developer & UX Architect

Sep 2020 - Sept 2023

Haitong International Securities Group Ltd · One of China's largest investment banks Hong Kong, China / Vancouver (Remote)

- **Scalable Enterprise Foundations:** Solely designed an enterprise-grade React architecture from the ground up over 6 months — OAuth, RBAC, and granular permissions — adopted across 2 trading platforms and extended by additional teams to 4 more business systems without architectural changes; retained as architectural owner throughout.
- **Design System:** Built and maintained a **shared component library** in Storybook adopted across 5 platforms — cascading improvements automatically without per-platform updates.
- **Rapid Development Framework:** Reduced form development time by **75%** by delivering a process-driven, dynamic framework supporting **200+ configurable trading workflows**.
- **Engineering Leadership:** Led a team of **5 frontend engineers** to deliver full digitization of **200+ workflows** within a 2-year timeline — overseeing code reviews, architectural guidance, and end-to-end quality standards.

Stack: React, Material UI, REST APIs, PostgreSQL, Jest, Docker, Kubernetes, Jenkins, Git, Vue

Frontend Developer / UI Developer

Dec 2019 - Sep 2020

Serai Limited · HSBC's first non-banking tech venture · B2B trade platform

Hong Kong, China

- **Data-Informed Decision Making:** Developed advanced **D3.js data-visualization** modules that surfaced supplier analytics, enabling internal teams to interpret complex supply-chain metrics efficiently. Led 2 frontend engineers across both products.
- **High-Performance UI:** Integrated **REST and GraphQL APIs** to power data-intensive fintech dashboards, ensuring low-latency performance for complex operational workflows.

Stack: React, Redux, Material UI, D3.js, Node.js, REST APIs, GraphQL, PostgreSQL, MongoDB, Jest, Cypress, AWS

Senior Frontend Developer

Dec 2017 - Sep 2019

Fayfay.com Limited · Travel Ecommerce · MoUs with Da Nang, Hue & HCMC Tourism Boards

Hong Kong, China

- **Marketing Efficiency Boost:** Reduced marketing implementation time by **70%** by delivering a library of reusable campaign components and a custom internal CMS for rapid product launches.
- **Full-Cycle E-commerce Scaling:** Built and scaled a comprehensive e-commerce engine, including secure **Stripe payment integrations**, checkout flows, and complex promotion modules.

Stack: React, Redux, Node.js, REST APIs, PostgreSQL, MongoDB, Stripe, Google Analytics, AWS

Web Developer

Nov 2015 - Dec 2017

The Hong Kong Jockey Club · HK's largest taxpayer · millions of users across racing, lottery & sports betting

Hong Kong, China

Developed high-traffic e-commerce and campaign platforms (hkjc.com) using ASP.NET and modern JavaScript.

Senior Web Developer

Sep 2014 - Oct 2015

Isobar Hong Kong Limited · APAC Digital Agency of the Year · Peninsula Hotels · Audi · UNIQLO

Hong Kong, China

Delivered high-profile digital campaigns for global brands including **Audi** and **UNIQLO**; managed junior developers.

Education

Master's in E-commerce,

The Hong Kong Polytechnic University (2019)

Bachelor of Arts in Applied and Media Arts,

The Hong Kong Polytechnic University (2013)

Certificate

Google **AI Professional** Certificate

Google **UX Design Professional** Certificate

Google **Digital Marketing & E-commerce** Certificate